University management app using python

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class Person:
university_name = "Codegnan University"  # Class Attribute (Shared by all
instances)

def __init__(self, name, age):
    self.name = name  # Instance Attribute
    self.age = age

def display_info(self):
    """Polymorphism: To be overridden by subclasses"""
    pass

class Student(Person):
    student_count = 0  # Class attribute for student count

    def __init__(self, name, age, student_id, course):
        super().__init__(name, age)  # Inheritance: Calling Parent Class Constructor
        self.__student_id = student_id  # Encapsulation: Private Attribute
        self.course = course
        Student.student_count += 1

    def display_info(self):
        """Polymorphism: Display student info"""
        print(f"Student: {self.name}, Age: {self.age}, Course: {self.course}")

    def get_student_id(self):
        """Encapsulation: Getter for private attribute"""
        return self.__student_id

class Faculty(Person):
    faculty_count = 0  # Class attribute for faculty count

    def __init__(self, name, age, faculty_id, department):
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        super().__init__(name, age)
        self.__faculty_id = faculty_id  # Encapsulation: Private Attribute
        self.department = department
        Faculty.faculty_count += 1

    def display_info(self):
        """Polymorphism: Display faculty info"""
        print(f"Faculty: {self.name}, Age: {self.age}, Department: {self.department}")

    @staticmethod
    def university_policy():
        """Static Method: Shared Information"""
        print("Codegnan University follows strict academic policies.")

    @classmethod
    def total_faculty(cls):
        """Class Method: Display total faculty count"""
        print(f"Total Faculty Members: {cls.faculty_count}")

# Creating Objects
student1 = Student("Rahul Sharma", 21, "CNU12345", "Computer Science")
student2 = Student("Ananya Reddy", 22, "CNU67890", "Data Science")

faculty1 = Faculty("Dr. Ravi Kumar", 45, "F001", "AI & ML")
faculty2 = Faculty("Dr. Meera Srinivas", 50, "F002", "Cybersecurity")

# Display Information
student1.display_info()
student2.display_info()
print(f"Student ID (Private Attribute): {student1.get_student_id()}")

faculty1.display_info()
faculty2.display_info()

# Calling Static and Class Methods
Faculty.university_policy()
Faculty.total_faculty()
```